Kepler’s laws

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use orbital shape, equal areas and period comparisons.

SUCCESS CRITERIA - I can...

1. Place the central body at a focus of an ellipse.
2. Compare speed near and far from the central body.
3. Use $T^2=a^3$ only with the specified units and central body.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A launch has v_x=**5 m/s** and v_y, i=**30 m/s** upward.
1. Use g=**10 m/s²**.
2. **Find time** to the top and total time back to launch height.

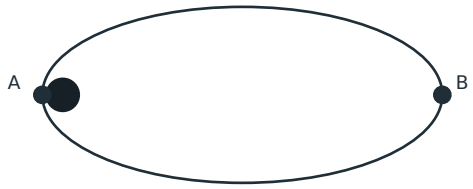
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $T^2 = a^3$.
For Sun orbits only when T is in years and a is in AU. a is the semi-major axis. Equal areas are swept in equal times.

Worked reasoning: For an ideal Sun orbit, a=**4 AU** gives T=√(**4³**)=**8 years**. The orbit period depends on semimajor axis, not on a chosen point's instantaneous speed alone.

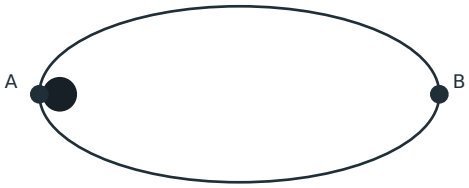
Your turn: Draw an ellipse, put the Sun at a focus and **mark the faster** and slower parts.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark the Sun at a focus on an ellipse. **Label where the object** is fastest and slowest.



Marzano check - circle: ① ② ③

04 Apply the physics

Supplied comparisons: a=**1 AU** gives T=**1 year**; a=**4 AU** gives T=**8 years**. Which orbit takes longer? **How many** Earth years for two orbits of the outer object?

Semimajor axis (AU)	Period (years)
1	1
4	8

Marzano check - circle: ① ② ③

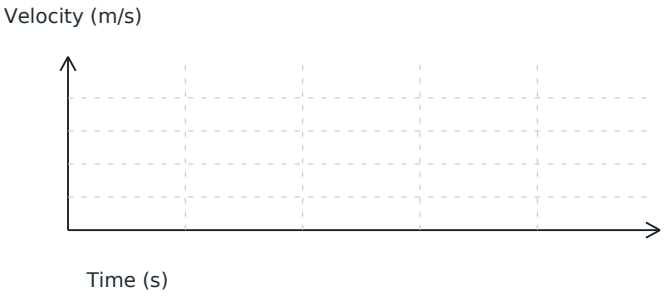
05 Explain your reasoning

Does knowing an orbit’s period alone tell the date when an asteroid reaches Earth’s location? **What else** is needed?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

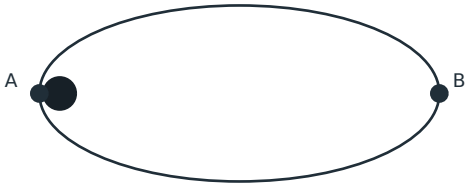
A ball is dropped from rest and falls ideally for **5 s**. Use g=**10 m/s²**. **Find its speed** and direction.
Sketch the motion and **label a positive direction**.



Marzano check - circle: ① ② ③

07 Exit check

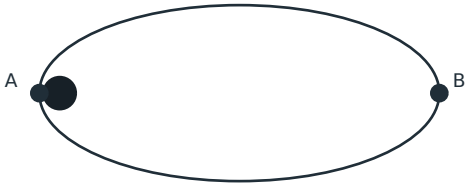
An asteroid is nearest the Sun at point A and farthest at B. At which point is it fastest? **Name which law helps explain** this.



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Two Sun orbits have supplied periods **1** and **8 years**. Their semimajor axes are **1** and **4 AU**. **Explain why four times the** size does not mean four times the period.



Marzano check - circle: ① ② ③